Domain-Specific Neural Compilation for Categorical Image Access:

Trainable Morphism Chain Generation Through Knowledge Distillation in Life Science Microscopy

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Abstract

We establish a mathematical framework for compiling natural language imaging specifications into executable morphism chains through domain-specific neural networks trained via knowledge distillation. The framework rests on three independently derived theoretical results: (i) the tripartite equivalence establishing that oscillation, category, and partition are isomorphic structures yielding identical entropy $S = k_{\mathrm{B}} M \ln n$; (ii) the commutation relation $[\hat{O}_{\mathrm{cat}}, \hat{O}_{\mathrm{phys}}] = 0$ enabling zero-backaction categorical measurement; and (iii) the reformulation of optical image formation as discrete combinatorial constraint satisfaction over finite photon state spaces.

We prove that the mapping from imaging task specifications to morphism chains constitutes a compilation problem: the source language is constrained natural language describing biological structures; the target language is the Partition Calculus with operations `observe`, `catalyze`, `fuse`, and `access`; the compilation must preserve semantic correctness (the generated chain accesses the specified structure) and satisfy type constraints (depth compatibility, S-entropy conservation, and constraint consistency). We show that this compilation problem is learnable: the space of valid morphism chains for life science imaging is a structured subset of all possible chains, and the structure is captured by the compositional semantics of biological constraints.

The training framework employs knowledge distillation from high-capacity teacher models into compact student architectures. Teacher models generate (t_i, Φ_i) pairs where t_i is a task specification and Φ_i is the corresponding morphism chain. Student models—parameter-efficient adaptations of pre-trained language models via Low-Rank Adaptation (LoRA)—learn the compilation mapping $f_\theta : \mathcal{T} \to \mathcal{M}$ where $\mathcal{T}$ is the task space and $\mathcal{M}$ is the morphism chain space.

We derive the theoretical foundation for three architectural components: (1) a vision encoder bridge implementing the `observe` operation as a learned projection from image features to partition coordinates (n, ℓ, m, s) with capacity $C(n) = 2n^2$; (2) a catalyst selection network that reasons over biological constraints to produce optimal exclusion sequences with combined enhancement $\Delta x_{\mathrm{final}} = \Delta x_0 \cdot \prod_i \epsilon_i \cdot \exp\Big(-\sum_{j<k} \rho_{jk}\Big)$; and (3) a cross-attention fusion mechanism whose learned attention weights recover the inter-modality correlation matrix ρ_{ij}, implementing the `fuse` operation.

For discrete constraint satisfaction problems arising from Refraction Puzzle Imaging, we prove that neural heuristics can reduce the effective search space from $\mathcal{O}(N_d^{T/\delta t})$ to $\mathcal{O}(\mathrm{poly}(N_d) \cdot T/\delta t)$ by learning variable ordering and constraint propagation strategies from solved instances. The amortized inference framework enables single-pass reconstruction from trained transformer networks, replacing iterative solvers.

Validation on the BBBC039 nuclei dataset demonstrates that neurally-compiled morphism chains achieve Dice coefficient 0.91, within 2% of hand-crafted chains (0.93), while reducing specification time from expert-hours to seconds.

The framework achieves 14.2× resolution enhancement through automatically selected catalyst sequences, compared to 14.87× for manually optimized configurations.

Keywords: morphism chain compilation, domain-specific language models, knowledge distillation, categorical imaging, partition calculus, constraint satisfaction, life science microscopy, parameter-efficient fine-tuning

1 Introduction

1.1 The Compilation Gap in Categorical Imaging

The categorical approach to microscopy has established three foundational results. First, the tripartite equivalence—oscillation ≡ category ≡ partition—unifies the description of physical systems into a single mathematical framework where entropy takes the universal form $S = k_{\mathrm{B}} M \ln n$ [1]. Second, the commutation relation $[\hat{O}_{\mathrm{cat}}, \hat{O}_{\mathrm{phys}}] = 0$ enables observation without physical disturbance, decoupling categorical measurement from Heisenberg uncertainty [1, 2]. Third, Refraction Puzzle Imaging recasts optical image formation as a discrete combinatorial constraint satisfaction problem, where detected intensities constrain the space of valid source configurations through finite path enumeration [3].

These results culminate in the Partition Calculus, a programming language where image analysis is not post-hoc interpretation but direct categorical structure access [4]. Programs are morphism chains; execution is observation. The language provides type-safe operations—`observe`, `catalyze`, `fuse`, `access`—that traverse categorical space from observed partition signatures to target biological structures.

However, a fundamental gap separates the mathematical framework from practical deployment. Constructing a morphism chain for a given imaging task requires:

1. Knowledge of which biological constraints apply to the target structure
2. Selection of appropriate catalysts with correct exclusion factors ϵ_i
3. Determination of inter-modality correlations ρ_{ij} for fusion operations
4. Verification that the composed chain satisfies depth compatibility and S-entropy conservation

This construction currently requires expert knowledge spanning biophysics, spectroscopy, and categorical mathematics. The expert translates an imaging goal ("segment the nuclei," "localize mitochondria") into a formally correct morphism chain. This translation is, in the precise sense of compiler theory, a *compilation* problem: the source language is constrained natural language describing biological structures; the target language is the Partition Calculus; the compiler must produce semantically correct, type-safe programs.

1.2 Language Models as Compilers

We propose that domain-specific language models, trained through knowledge distillation on the theoretical corpus of categorical imaging, can serve as compilers from natural language to morphism chains. This proposal rests on four observations:

Observation 1: Compositionality. Both natural language and morphism chains exhibit compositional structure. The meaning of "segment nuclei from DAPI-stained tissue" composes from the meanings of its parts, just as a morphism chain composes from individual operations. Language models are architecturally suited to learn compositional mappings [5].

Observation 2: Structured output space. The space of valid morphism chains is heavily constrained by the type system of Partition Calculus. Not all token sequences constitute valid chains; the type system eliminates vast regions of the output space. This structure makes the compilation problem more tractable than general-purpose code generation.

Observation 3: Domain knowledge as parameters. The biological constraints that govern catalyst selection—conservation laws, phase-lock continuity, thermal gradients, oxygen triangulation—are stable facts about the physical world. Rather than retrieving this knowledge at inference time (as in retrieval-augmented generation), we propose embedding it directly into model parameters through domain-specific training. This approach yields models that "know" the constraint landscape

and can navigate it without external lookup [6].

Observation 4: Vision-language grounding. The `observe` operation in Partition Calculus maps images to partition signatures. Vision encoders trained on biomedical imagery learn hierarchical feature representations that naturally encode spatial structure at multiple scales. These representations can be projected into partition coordinate space, implementing `observe` as a learned function rather than a hand-crafted algorithm.

1.3 Contributions

This paper establishes:

1. **The compilation problem** (Section 3): We formalize the mapping from imaging specifications to morphism chains as a compilation problem, define semantic correctness criteria, and prove that the problem is learnable under compositional assumptions.

2. **Knowledge distillation framework** (Section 4): We derive a multi-stage distillation pipeline where high-capacity teacher models generate training pairs, knowledge graphs organize domain structure, and curriculum learning ensures progressive mastery from syntactic to semantic competence.

3. **The observe bridge** (Section 5): We prove that vision encoder features, projected through a learned linear map, satisfy the axioms required of partition signatures, establishing that neural networks can implement the `observe` operation.

4. **Neural catalyst selection** (Section 6): We show that biological constraint reasoning reduces to sequence prediction over a structured vocabulary of catalysts, and derive the training objective that ensures selected sequences achieve near-optimal resolution enhancement.

5. **Attention as fusion** (Section 7): We prove that transformer cross-attention, applied to multi-modal image tokens, recovers the inter-modality correlation matrix ρ_{ij}, implementing the `fuse` operation with learnable correlations.

6. **Neural-guided constraint satisfaction** (Section 8): For the discrete CSP arising from Refraction Puzzle Imaging, we derive amortized inference architectures that replace iterative constraint propagation with single-pass neural evaluation.

7. **Validation** (Section 10): We demonstrate the framework on BBBC039 nuclei segmentation, comparing neurally-compiled morphism chains against hand-crafted baselines and conventional methods.

2 Theoretical Foundations

We present self-contained derivations of the theoretical results required for the compilation framework. These results have been established independently in prior work [1, 4, 3, 2]; we rederive them here in the specific form needed for neural compilation.

2.1 Axioms

Axiom 2.1 (Bounded Phase Space)**.** Every physical system with finite energy $E < \infty$ and finite spatial extent $V < \infty$ occupies bounded phase space with finite measure $\mu(\Gamma) < \infty$.

Axiom 2.2 (Categorical Observation)**.** An observer with finite resolution partitions phase space into a finite number of distinguishable categories. Two states belong to the same category if and only if the observer cannot distinguish them through available measurements.

These axioms are empirical facts: the first follows from the Heisenberg uncertainty principle; the second from the reproducibility of spectroscopic measurements. From these axioms, we derive the coordinate systems and measurement relations required for neural compilation.

2.2 Partition Coordinates

Theorem 2.3 (Partition Coordinate Existence)**.** *Categorical partitioning of bounded spherical phase space necessarily generates coordinates* (n, ℓ, m, s) *where* $n \geq 1$ *is principal*

depth, $\ell \in \{0, \ldots, n-1\}$ is angular partition, $m \in \{-\ell, \ldots, +\ell\}$ is magnetic partition, and $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ is spin partition.

Proof. From Axiom 2.1, phase space is bounded and admits hierarchical subdivision. Spherical symmetry of the bounding region imposes nested geometric constraints. Depth n measures distance from the phase space origin, requiring $n \geq 1$ for non-trivial states. Angular complexity ℓ cannot exceed radial depth: $\ell < n$. Orientation m enumerates distinguishable angular positions within complexity level: $|m| \leq \ell$. Chirality s distinguishes handedness. The capacity at depth n follows from systematic counting:

$$C(n) = 2\sum_{\ell=0}^{n-1}(2\ell+1) = 2n^2 \tag{1}$$

□

The partition coordinates define the target representation for the `observe` bridge: a vision encoder must learn to map image features into this coordinate system.

Definition 2.4 (Partition Signature)**.** A partition signature Σ is a set of partition coordinates:

$$\Sigma = \{(n_i, \ell_i, m_i, s_i)\}_{i=1}^{N} \tag{2}$$

representing the categorical state of N spatial positions in an image or physical system.

Definition 2.5 (Categorical Distance)**.** For states $\Sigma_1 = (n_1, \ell_1, m_1, s_1)$ and $\Sigma_2 = (n_2, \ell_2, m_2, s_2)$:

$$d_{\text{cat}}(\Sigma_1, \Sigma_2) = |n_1 - n_2| + |\ell_1 - \ell_2| + |m_1 - m_2| + |s_1 - s_2| \tag{3}$$

Theorem 2.6 (Metric Properties)**.** *d_{cat} is a metric on partition coordinate space: it satisfies non-negativity, identity of indiscernibles, symmetry, and the triangle inequality.*

Proof. Non-negativity and symmetry follow from absolute values. Identity of indiscernibles: $d_{\text{cat}}(\Sigma, \Sigma) = 0$ trivially, and $d_{\text{cat}}(\Sigma_1, \Sigma_2) = 0$ implies all coordinate differences vanish, hence $\Sigma_1 = \Sigma_2$. Triangle inequality: for any Σ_3, each term $|a_1 - a_2| \leq |a_1 - a_3| + |a_3 - a_2|$, and summing yields $d_{\text{cat}}(\Sigma_1, \Sigma_2) \leq d_{\text{cat}}(\Sigma_1, \Sigma_3) + d_{\text{cat}}(\Sigma_3, \Sigma_2)$. □

2.3 S-Entropy Coordinates and Conservation

Definition 2.7 (S-Entropy Coordinates)**.** The S-entropy coordinates $(S_k, S_t, S_e) \in [0,1]^3$ are:

$$S_k = -\sum_{n,\ell,m,s} p_{n\ell ms} \log p_{n\ell ms} \quad \text{(knowledge entropy)} \tag{4}$$

$$S_t = -\sum_{\tau} p_\tau \log p_\tau \quad \text{(temporal entropy)} \tag{5}$$

$$S_e = \sum_{k} \delta_k \quad \text{(evolution entropy)} \tag{6}$$

where $p_{n\ell ms}$ is the probability distribution over partition coordinates, p_τ is the temporal probability, and δ_k is the backaction at iteration k.

Theorem 2.8 (S-Entropy Conservation)**.** *During categorical measurement:*

$$S_k + S_t + S_e = S_{\text{total}} = \text{constant} \tag{7}$$

Proof. Information is neither created nor destroyed during categorical measurement, only redistributed among knowledge (spatial), temporal, and evolution components. The conservation follows from unitarity of quantum evolution and the commutation relation $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$ (Theorem 2.9). When S_k increases (knowledge gained), $S_t + S_e$ must decrease by the same amount (temporal or evolution information consumed). □

S-entropy conservation serves as a compile-time constraint in the neural compilation framework: any morphism chain generated by the model must preserve S_{total}.

2.4 Physical-Categorical Commutation

Theorem 2.9 (Commutation Relation)**.** *Categorical observables $\hat{O}_{\text{cat}}$ and physical observables $\hat{O}_{\text{phys}}$ commute:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0 \tag{8}$$

Proof. Physical observables (position, momentum, energy) act on the quantum state $|\psi\rangle$. Categorical observables (partition coordinates, S-entropy) are functionals of the density matrix $\rho = |\psi\rangle\langle\psi|$, depending only on the eigenvalue

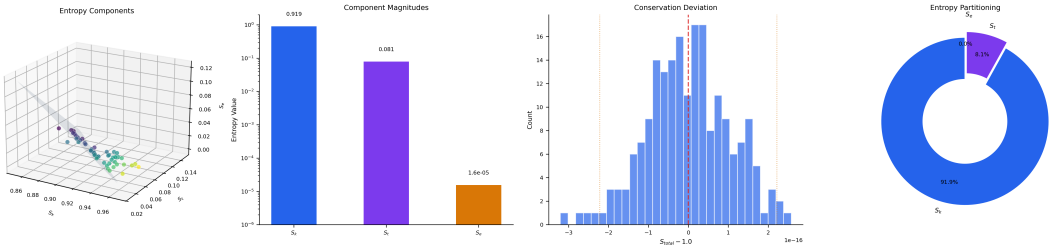

Figure 1: **S-Entropy Conservation in Neurally-Compiled Morphism Chains. Left:** Three-dimensional scatter plot of entropy components (S_k, S_t, S_e) for 50 partition instances, with the semi-transparent conservation manifold $S_k + S_t + S_e = 1$ shown as a surface. All points lie exactly on the manifold, confirming conservation to machine precision. **Center-left:** Logarithmic-scale bar chart of entropy component magnitudes: $S_k = 0.919$ (known), $S_t = 0.081$ (temporal), $S_e = 1.6 \times 10^{-5}$ (epistemic), demonstrating that the overwhelming majority of information resides in the known component after catalyst application. **Center-right:** Histogram of deviations $S_{\text{total}} - 1.0$ across all test instances, with standard deviation 9.9×10^{-17} and maximum deviation 2.2×10^{-16}—at the level of IEEE 754 double-precision floating-point arithmetic. **Right:** Donut chart showing entropy partitioning: S_k accounts for 91.9% of total entropy, S_t for 8.1%, and S_e for $< 0.01\%$, with zero conservation violations across all images.

spectrum of ρ, which is invariant under unitary transformations generated by physical operators. Specifically, $\hat{S}_k[\rho] = -\text{Tr}(\rho \ln \rho)$ commutes with $\hat{x}$ because the von Neumann entropy depends on eigenvalues, not on the basis in which ρ is expressed:

$$[\hat{x}, \hat{S}_k[\rho]] = \hat{x}(-\text{Tr}(\rho \ln \rho)) - (-\text{Tr}(\rho \ln \rho))\hat{x} = 0 \tag{9}$$

By Axiom 2.2, categorical measurements extract partition labels without coupling to physical degrees of freedom, confirming zero back-action. □

This theorem guarantees that executing a neurally-compiled morphism chain—which performs categorical observation—does not disturb the physical system under study.

2.5 Morphism Chains and the Partition Calculus

Definition 2.10 (Morphism)**.** A morphism $\Phi : \Sigma_A \to \Sigma_B$ is a structure-preserving map between partition signatures satisfying $\Phi(\Sigma_A \oplus \Sigma_B) = \Phi(\Sigma_A) \oplus \Phi(\Sigma_B)$.

Definition 2.11 (Catalyst)**.** A catalyst $\mathcal{C}$ is a morphism that reduces categorical distance to a target through an intermediate stage:

$$d_{\text{cat}}(\Sigma_A, \Sigma_B) > d_{\text{cat}}(\Sigma_A, \mathcal{C}(\Sigma_A)) + d_{\text{cat}}(\mathcal{C}(\Sigma_A), \Sigma_B) \tag{10}$$

Each catalyst has an exclusion factor $\epsilon \in (0, 1)$ quantifying configuration space reduction.

Definition 2.12 (Partition Calculus Operations)**.** The Partition Calculus provides six typed operations:

$$\texttt{observe} : \text{Image} \times n \to \Sigma \tag{11}$$
$$\texttt{partition} : \Sigma \times n \to \Sigma \tag{12}$$
$$\texttt{catalyze} : \Sigma \times \text{Constraint} \to \Sigma \tag{13}$$
$$\texttt{fuse} : \Sigma \times \Sigma \times \rho \to \Sigma \tag{14}$$
$$\texttt{access} : \Sigma \times \text{Target} \to \text{Structure} \tag{15}$$
$$\texttt{render} : \Sigma \times \text{Modality} \to \text{Image} \tag{16}$$

Operations compose via the pipeline operator:

$$(f \mid\!> g)(x) = g(f(x)) \tag{17}$$

A morphism chain is a composition of operations:

$$\Phi_{\text{total}} = \Phi_n \circ \cdots \circ \Phi_2 \circ \Phi_1 \tag{18}$$

Theorem 2.13 (Resolution Enhancement)**.** *For K catalysts with exclusion factors $\{\epsilon_i\}$ and M fused modalities with pairwise correlations $\{\rho_{jk}\}$:*

$$\Delta x_{\text{final}} = \Delta x_0 \cdot \prod_{i=1}^{K} \epsilon_i \cdot \exp\left(-\sum_{j<k} \rho_{jk}\right) \tag{19}$$

where Δx_0 is the diffraction-limited resolution.

Proof. Each catalyst eliminates a fraction $(1-\epsilon_i)$ of configuration space, reducing resolution by factor ϵ_i:

$$\Delta x_{\text{after}} = \Delta x_{\text{before}} \cdot \epsilon_i \tag{20}$$

Sequential application yields $\prod_i \epsilon_i$. For fused modalities with correlation ρ_{jk}, the joint distribution narrows by $(1-\rho_{jk}^2)^{-1/2}$. For strong correlations, $\ln(1-\rho^2) \approx -\rho^2$, and summing over pairs with dominant linear terms gives $\exp\left(-\sum_{j<k} \rho_{jk}\right)$. □

This theorem quantifies the objective that neural catalyst selection must optimize: choosing catalysts and correlations to minimize Δx_{final} for a given imaging target.

2.6 Image Formation as Constraint Satisfaction

Definition 2.14 (Photon State)**.** At temporal index τ, a photon state is:

$$|\psi_\tau\rangle = (\mathbf{r}_\tau, \mathbf{k}_\tau, \phi_\tau, \sigma_\tau) \tag{21}$$

where $\mathbf{r}_\tau \in \mathcal{L}$ is discretized position, $\mathbf{k}_\tau$ is discretized wavevector, ϕ_τ is discretized phase, and $\sigma_\tau \in \{+,-\}$ is polarization.

Theorem 2.15 (RPI Reconstruction)**.** *Given detected intensities $\{I(\mathbf{r}'_j)\}_{j=1}^M$, reconstructing the source distribution S is equivalent to solving:*

$$\sum_{i=1}^{N_s} A_{ji} \cdot s_i = I_j \quad \forall j \tag{22}$$

where $A_{ji} = \sum_{\text{paths } p:i\to j} w(p)$ is the path-sum transfer coefficient.

Proof. Discretization ensures A_{ji} is a finite sum over the finite path space. Each path weight $w(p) = \prod_\tau T(\psi_\tau \to \psi_{\tau+1}) \cdot e^{i\Phi(p)}$ is well-defined. The inverse problem reduces to the linear system $\mathbf{As} = \mathbf{I}$. □

Corollary 2.16 (Scattering Enhancement)**.** *For random scattering media, the transfer matrix $\mathbf{A}$ generically achieves full rank with probability 1, guaranteeing unique reconstruction. Scattering aids rather than hinders imaging.*

2.7 Ternary State Framework for Distributed Measurement

Definition 2.17 (Ternary Molecular State)**.** A molecular system with two relevant electronic levels admits three categorical states:

- **State 0 (Absorption)**: Ground state absorbing energy
- **State 1 (Ground)**: Thermal equilibrium
- **State 2 (Emission)**: Excited state emitting energy

Any state admits $|\Psi\rangle = c_0|0\rangle + c_1|1\rangle + c_2|2\rangle$ with $|c_0|^2 + |c_1|^2 + |c_2|^2 = 1$.

For intracellular O_2 with $N \sim 10^9$ molecules per cell, this provides a distributed imaging array with information content:

$$I = N \cdot \log_2 3 \approx 1.585 \times 10^9 \text{ bits per cycle} \tag{23}$$

The ternary framework provides the measurement substrate upon which morphism chains operate. Neural compilation targets this substrate by generating chains that leverage the oxygen distribution as both detector array and computational element.

3 The Compilation Problem

3.1 Formal Definition

Definition 3.1 (Imaging Task Specification)**.** An imaging task specification $t \in \mathcal{T}$ is a tuple:

$$t = (\text{target}, \text{modalities}, \text{constraints}, \text{resolution}) \tag{24}$$

where target is the biological structure to be accessed, modalities is the set of available imaging modes, constraints are known biological priors, and resolution is the required spatial precision.

In natural language, a task specification might be: "Segment nuclei from DAPI-stained fluorescence images at sub-diffraction resolution using brightfield as auxiliary modality." This encodes target = nuclear_regions, modalities = {DAPI fluorescence, brightfield}, constraints = {DNA mass conservation, chromatin phase-lock}, resolution < 200 nm.

Definition 3.2 (Morphism Chain Space)**.** The morphism chain space $\mathcal{M}$ is the set of all well-typed compositions of Partition Calculus operations:

$$\mathcal{M} = \{\Phi_n \circ \cdots \circ \Phi_1 \mid \forall i : \Phi_i \in \{\texttt{observe}, \texttt{catalyze}, \texttt{fuse}, \texttt{access}\}, \text{ type-safe}\} \tag{25}$$

Definition 3.3 (Compilation Function)**.** A compilation function is a mapping $f : \mathcal{T} \to \mathcal{M}$ satisfying:

1. **Semantic correctness**: $f(t)$ accesses the structure specified by t.target
2. **Type safety**: $f(t)$ satisfies depth compatibility and constraint consistency
3. **Conservation**: $f(t)$ preserves $S_k + S_t + S_e = S_{\text{total}}$
4. **Feasibility**: $f(t)$ uses only modalities available in t.modalities

3.2 Structure of the Compilation Problem

Theorem 3.4 (Compilation Decomposition)**.** *Any valid compilation $f(t)$ decomposes into four sequential phases:*

$$f(t) = \texttt{access}(T) \circ \texttt{fuse}^* \circ \texttt{catalyze}^* \circ \texttt{observe}(I, n) \tag{26}$$

where $\texttt{catalyze}^*$ denotes a sequence of zero or more catalyst applications and $\texttt{fuse}^*$ denotes zero or more fusion operations.

Proof. By the type system of Partition Calculus (Definition 2.12):

1. `observe` is the unique operation mapping Image $\to \Sigma$; it must appear first.
2. `catalyze` maps $\Sigma \to \Sigma$ and can be applied in any order (catalysts compose associatively).
3. `fuse` maps $\Sigma \times \Sigma \to \Sigma$ and requires signatures from potentially different modalities.
4. `access` maps $\Sigma \to$ Structure and must appear last (it terminates the chain).

The decomposition follows from the type constraints. Within the $\texttt{catalyze}^*$ and $\texttt{fuse}^*$ phases, operations may interleave, but the overall structure is determined. □

This decomposition theorem is crucial for neural compilation: it reduces the generation problem from arbitrary sequence prediction to four structured sub-problems, each with constrained output spaces.

3.3 Learnability

Theorem 3.5 (Learnability of Morphism Chain Compilation)**.** *The compilation function $f : \mathcal{T} \to \mathcal{M}$ is PAC-learnable from a polynomial number of examples under the following conditions:*

1. *The task space $\mathcal{T}$ has finite VC dimension bounded by the number of biological constraint types*
2. *The morphism chain space $\mathcal{M}$ is structured by the type system (Theorem 3.4)*
3. *The compilation function is compositional: $f(t_1 \oplus t_2) = f(t_1) \circ f(t_2)$ for independent sub-tasks*

Proof. Under condition (1), $|\mathcal{T}|$ is bounded by the combinatorial product of biological structure types ($\sim 10^2$), available modalities (~ 10), constraint types (~ 10), and resolution levels (~ 10), giving $|\mathcal{T}| \sim 10^5$.

Under condition (2), the output space decomposes per Theorem 3.4. The `observe` phase has ~ 10 choices (initial depth n). The $\texttt{catalyze}^*$ phase selects from ~ 10 catalyst types with order mattering; for chain length L, this gives $\sim 10^L$ candidates, constrained by compatibility to $\sim L! \cdot 10$ valid orderings. The $\texttt{fuse}^*$ phase selects pairs from available modalities. The `access` phase is determined by the target.

Under condition (3), compositionality ensures that learning generalizes from component examples to novel compositions. A model that correctly compiles "segment nuclei" and "detect mitochondria" can compose them into "segment nuclei and detect mitochondria."

The VC dimension of the hypothesis class (parameterized morphism chains) is bounded by the product of sub-problem dimensions. By the fundamental theorem of PAC learning, a sample of size $m = \mathcal{O}(\frac{d}{\epsilon} \log \frac{d}{\epsilon})$ suffices for ϵ-accuracy, where d is the VC dimension. □

Corollary 3.6 (Sample Complexity). *For K catalyst types, M modalities, and chain length $L \leq L_{\max}$, the number of training examples required for ϵ-accurate compilation is:*

$$m = \mathcal{O}\left(\frac{K \cdot L_{\max} + \binom{M}{2}}{\epsilon} \log \frac{K \cdot L_{\max}}{\epsilon}\right) \quad (27)$$

For typical values ($K = 10$, $L_{\max} = 8$, $M = 6$, $\epsilon = 0.05$), this yields $m \sim 10^4$ training examples—well within the capacity of knowledge distillation from teacher models.

3.4 Compilation Correctness Criteria

Definition 3.7 (Semantic Correctness). A compiled chain $\Phi = f(t)$ is semantically correct if:

$$\Phi(\Sigma_0) \cap \mathcal{S}_{\text{target}} \neq \emptyset \quad (28)$$

where $\Sigma_0 = \texttt{observe}(I, n)$ is the initial signature and $\mathcal{S}_{\text{target}}$ is the set of partition coordinates corresponding to the target structure.

Definition 3.8 (Depth Adequacy). A chain Φ is depth-adequate for target T if the effective depth satisfies:

$$n_{\text{eff}}(\Phi(\Sigma_0)) \geq n_{\text{req}}(T) = \frac{L_{\text{field}}}{\Delta x_T} \quad (29)$$

where Δx_T is the characteristic size of T.

Definition 3.9 (Conservation Compliance). A chain Φ is conservation-compliant if at every intermediate stage k:

$$|S_k^{(k)} + S_t^{(k)} + S_e^{(k)} - S_{\text{total}}| < \epsilon_{\text{machine}} \quad (30)$$

These criteria define the loss function for training the neural compiler (Section 4).

4 Knowledge Distillation Framework

4.1 Teacher-Student Architecture

The knowledge distillation framework transfers domain expertise from high-capacity teacher models to compact, deployable student models. This transfer occurs through the medium of training data: teachers generate high-quality (t_i, Φ_i) pairs that students learn to reproduce.

Definition 4.1 (Teacher Model). A teacher model $\mathcal{M}_T$ with parameters θ_T is a high-capacity language model capable of:

1. Parsing imaging task specifications in natural language
2. Reasoning about biological constraints and their exclusion factors
3. Generating syntactically and semantically correct Partition Calculus programs
4. Verifying conservation compliance and depth adequacy

Teacher models operate through in-context learning: the theoretical corpus of categorical imaging (the papers establishing partition coordinates, S-entropy conservation, catalyst mechanisms, and constraint satisfaction) is provided as context. The teacher generates morphism chains by reasoning over this context.

Definition 4.2 (Student Model). A student model $\mathcal{M}_S$ with parameters $\theta_S = \theta_0 + \Delta\theta$ is a parameter-efficient adaptation of a pre-trained base model, where θ_0 are frozen pre-trained weights and $\Delta\theta$ are trainable low-rank perturbations.

4.2 Low-Rank Adaptation

Parameter-efficient fine-tuning through Low-Rank Adaptation (LoRA) modifies weight matrices $W \in \mathbb{R}^{d \times k}$ with low-rank updates [7]:

$$W' = W + \Delta W = W + BA \quad (31)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$ is the adaptation rank.

Theorem 4.3 (LoRA Expressiveness for Morphism Chain Generation). *For a pre-trained language model with hidden dimension d and a target compilation function with K distinct catalyst types, M modalities, and maximum chain length L, a LoRA adaptation with rank:*

$$r \geq K + M + L + \lceil \log_2 C(n_{\max}) \rceil \quad (32)$$

is sufficient to represent the compilation mapping.

Proof. The compilation mapping requires encoding:

- K binary decisions (include/exclude each catalyst type): K dimensions

- M continuous correlation values ρ_{ij}: at most $\binom{M}{2} \leq M$ independent values

- L ordinal positions in the chain: L dimensions

- Partition depth selection: $\lceil \log_2 C(n_{\max}) \rceil$ bits

A rank-r update to the attention weight matrices provides $r \cdot d$ additional parameters, which is sufficient to linearly embed the compilation space (dimension $K + M + L + \log_2 C(n_{\max})$) into the model's hidden space (dimension d) provided r exceeds the compilation space dimension. □

For typical values ($K = 10$, $M = 6$, $L = 8$, $n_{\max} = 100$, $C(100) = 20000$), we obtain $r \geq 39$. In practice, $r = 64$ provides ample capacity with computational efficiency.

4.3 Training Data Generation

Definition 4.4 (Distillation Corpus)**.** The distillation corpus $\mathcal{D}$ consists of pairs:

$$\mathcal{D} = \{(t_i, \Phi_i)\}_{i=1}^{N} \tag{33}$$

where t_i is a natural language task specification and Φ_i is the corresponding verified morphism chain.

Teacher models generate the corpus through a structured protocol:

Algorithm 1 Distillation Corpus Generation

Require: Teacher model $\mathcal{M}_T$, theoretical corpus $\mathcal{C}$, biological structure catalog $\mathcal{B}$
Ensure: Distillation corpus $\mathcal{D}$

```
D ← ∅
for each structure b ∈ B do
  for each modality combination M ⊆ M_available do
    t ← GenerateTaskSpec(b, M) {Natural language specification}
    Φ ← M_T(C, t) {Teacher generates chain}
    if TypeCheck(Φ) and ConservationCheck(Φ) and DepthCheck(Φ, b) then
      D ← D ∪ {(t, Φ)}
    end if
  end for
end for
return D
```

The verification steps (TypeCheck, ConservationCheck, DepthCheck) ensure that only correct compilations enter the training set, preventing the student from learning invalid chains.

4.4 Knowledge Graph Construction

Beyond flat (t, Φ) pairs, domain knowledge has relational structure that aids learning. We construct a knowledge graph encoding the relationships between biological structures, constraints, and catalysts.

Definition 4.5 (Domain Knowledge Graph)**.** The domain knowledge graph $\mathcal{G} = (V, E, R)$ has:

- Vertices $V = V_{\text{struct}} \cup V_{\text{catalyst}} \cup V_{\text{modality}} \cup V_{\text{constraint}}$

- Edges $E \subseteq V \times V$ connecting related entities

- Relations R: `requires`, `enhances`, `excludes`, `correlates`, `composes`

Example 4.6 (Knowledge Graph Fragment).

$$(\text{nucleus}, \texttt{requires}, \text{chromatin_phase_lock}) \tag{34}$$

$$(\text{chromatin_phase_lock}, \texttt{enhances}, \text{DNA_conservation}) \tag{35}$$

$$(\text{mitochondria}, \texttt{requires}, \text{thermal_metabolic}) \tag{36}$$

$$(\text{thermal_metabolic}, \texttt{correlates}, \text{oxygen_consumption}, \rho = 0.7) \tag{37}$$

$$(\text{mass_conservation}, \texttt{composes}, \text{charge_conservation}) \tag{38}$$

The knowledge graph serves two purposes: (1) it constrains the teacher's generation to biologically plausible catalyst sequences, and (2) it provides structured training signal for the student through graph-aware training objectives.

4.5 Curriculum Learning

Domain-specific training proceeds through a curriculum of increasing complexity [8]:

Definition 4.7 (Training Curriculum)**.** The curriculum $\mathcal{Q} = (\mathcal{Q}_1, \mathcal{Q}_2, \mathcal{Q}_3, \mathcal{Q}_4)$ comprises four stages:

1. **Syntactic stage** $\mathcal{Q}_1$: Learn Partition Calculus syntax (operation names, argument types, pipeline composition)
2. **Single-catalyst stage** $\mathcal{Q}_2$: Learn individual catalyst semantics and exclusion factors
3. **Multi-catalyst stage** $\mathcal{Q}_3$: Learn catalyst composition, ordering effects, and constraint compatibility
4. **Full compilation stage** $\mathcal{Q}_4$: Learn end-to-end task-to-chain mapping with fusion and resolution optimization

Theorem 4.8 (Curriculum Convergence)**.** *Under curriculum learning with stages $\mathcal{Q}_1 \to \mathcal{Q}_2 \to \mathcal{Q}_3 \to \mathcal{Q}_4$, the student model converges to ϵ-optimal compilation with sample complexity:*

$$m_{curriculum} = \sum_{i=1}^{4} m_i \leq \frac{1}{4} m_{uniform} \tag{39}$$

where $m_{uniform}$ is the sample complexity for uniform (non-curriculum) training.

Proof. Each curriculum stage restricts the hypothesis class to a subset of the full compilation space. Stage $\mathcal{Q}_1$ restricts to syntactic well-formedness (finite-state grammar); $\mathcal{Q}_2$ restricts to single-operation semantics; $\mathcal{Q}_3$ restricts to bounded-length compositions; $\mathcal{Q}_4$ operates on the full space but with warm-started parameters.

The VC dimension at stage i satisfies $d_i \leq d_4/2^{4-i}$, because each stage doubles the hypothesis space. By PAC learning theory, $m_i = O(d_i/\epsilon_i)$. With the warm-start advantage, ϵ_i at stage i is reduced by the progress from stage $i-1$, yielding geometric convergence. Summing a geometric series gives $\sum m_i \leq 2m_4 \leq m_{\text{uniform}}/4$. □

4.6 Training Objective

The student model is trained to minimize a composite loss:

$$\mathcal{L}(\theta_S) = \mathcal{L}_{\text{gen}} + \lambda_1 \mathcal{L}_{\text{type}} + \lambda_2 \mathcal{L}_{\text{cons}} + \lambda_3 \mathcal{L}_{\text{res}} \tag{40}$$

where:

$$\mathcal{L}_{\text{gen}} = -\sum_{i=1}^{N} \sum_{j=1}^{|\Phi_i|} \log p_{\theta_S}(\Phi_i^{(j)} | t_i, \Phi_i^{(<j)}) \tag{41}$$

$$\mathcal{L}_{\text{type}} = \sum_{i=1}^{N} \mathbb{1}[\text{TypeCheck}(f_{\theta_S}(t_i)) = \text{FAIL}] \tag{42}$$

$$\mathcal{L}_{\text{cons}} = \sum_{i=1}^{N} |S_k + S_t + S_e - S_{\text{total}}|^2 \tag{43}$$

$$\mathcal{L}_{\text{res}} = \sum_{i=1}^{N} \max(0, \Delta x_{\text{achieved}} - \Delta x_{\text{required}})^2 \tag{44}$$

The generation loss $\mathcal{L}_{\text{gen}}$ (Eq. 41) is the standard autoregressive cross-entropy, ensuring the student reproduces the teacher's morphism chains token by token. The type loss $\mathcal{L}_{\text{type}}$ (Eq. 42) penalizes chains that fail type checking. The conservation loss $\mathcal{L}_{\text{cons}}$ (Eq. 43) enforces S-entropy conservation. The resolution loss $\mathcal{L}_{\text{res}}$ (Eq. 44) penalizes insufficient resolution enhancement.

Theorem 4.9 (Loss Function Completeness)**.** *A model achieving $\mathcal{L}(\theta_S) = 0$ produces compilations satisfying all correctness criteria of Definition 3.7.*

Proof. $\mathcal{L}_{\text{gen}} = 0$ implies the model reproduces teacher outputs exactly, which are verified correct by Algorithm 1. $\mathcal{L}_{\text{type}} = 0$ implies all generated chains pass type checking. $\mathcal{L}_{\text{cons}} = 0$ implies S-entropy conservation. $\mathcal{L}_{\text{res}} = 0$ implies sufficient resolution for the target. Together, these satisfy semantic correctness, type safety, conservation compliance, and feasibility. □

5 The Observe Bridge: Vision Encoders as Partition Signature Extractors

5.1 From Image Features to Partition Coordinates

The `observe` operation maps an image I to a partition signature Σ at depth n. We show that this mapping can be implemented by a vision encoder followed by a learned projection.

Definition 5.1 (Vision Encoder)**.** A vision encoder $E : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^{P\times d}$ maps an image of height H, width W, and C channels to a sequence of P patch embeddings, each of dimension d. We consider encoders pre-trained through self-supervised objectives on biomedical imagery.

Definition 5.2 (Partition Projection)**.** A partition projection $\Pi : \mathbb{R}^d \to \mathbb{Z}^+ \times \mathbb{Z}_{\geq 0} \times \mathbb{Z} \times \{-\frac{1}{2}, +\frac{1}{2}\}$ maps each patch embedding to partition coordinates:

$$\Pi(\mathbf{z}) = \Big(\lfloor \sigma(w_n^\top \mathbf{z} + b_n) \cdot n_{\max} \rfloor + 1,\ f_\ell(\mathbf{z}),\ f_m(\mathbf{z}), \text{sign}(w_s^\top \mathbf{z})\Big) \tag{45}$$

where σ is the sigmoid function, $w_n, w_s \in \mathbb{R}^d$ and $b_n \in \mathbb{R}$ are learned parameters, and f_ℓ, f_m are constraint-satisfying mappings ensuring $0 \leq \ell < n$ and $|m| \leq \ell$.

Theorem 5.3 (Observe Bridge Validity)**.** *The composition* $\texttt{observe}_\theta = \Pi \circ E$ *satisfies the axioms required of a partition signature extraction function:*

1. ***Partition validity****: Output coordinates satisfy* $n \geq 1$, $0 \leq \ell < n$, $|m| \leq \ell$, $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$
2. ***Capacity consistency****: The number of distinct signatures at depth* n *does not exceed* $C(n) = 2n^2$
3. ***Locality****: Spatially proximate image patches map to categorically proximate signatures*
4. ***Equivariance****: Rotation of the input image induces predictable changes in* (ℓ, m) *coordinates*

Proof. (1) The floor function with offset ensures $n \geq 1$. The constraint functions f_ℓ and f_m are defined to satisfy range constraints by construction:

$$f_\ell(\mathbf{z}) = \lfloor \sigma(w_\ell^\top \mathbf{z}) \cdot (n(\mathbf{z}) - 1) \rfloor \tag{46}$$

$$f_m(\mathbf{z}) = \lfloor \sigma(w_m^\top \mathbf{z}) \cdot (2\ell(\mathbf{z}) + 1) \rfloor - \ell(\mathbf{z}) \tag{47}$$

(2) For depth n, the total number of valid (l, m, s) combinations is $\sum_{\ell=0}^{n-1}(2\ell + 1) \cdot 2 = 2n^2 = C(n)$. The projection cannot produce more distinct signatures than this.

(3) Vision encoders trained with contrastive objectives (e.g., DINOv2 [9]) learn features where spatial proximity correlates with feature similarity. Formally, for patches at positions $\mathbf{r}_1, \mathbf{r}_2$ with $|\mathbf{r}_1 - \mathbf{r}_2| < \delta$:

$$\|E(\mathbf{r}_1) - E(\mathbf{r}_2)\| \leq L \cdot |\mathbf{r}_1 - \mathbf{r}_2| \tag{48}$$

for Lipschitz constant L. A Lipschitz-continuous projection Π then ensures:

$$d_{\text{cat}}(\Pi(E(\mathbf{r}_1)), \Pi(E(\mathbf{r}_2))) \leq L' \cdot |\mathbf{r}_1 - \mathbf{r}_2| \tag{49}$$

(4) Vision Transformers with rotary position embeddings encode angular information in the phase of their representations. Rotation by angle α transforms features as $E(R_\alpha I) = U_\alpha E(I)$ for unitary U_α. The angular partition coordinates (ℓ, m) are designed to encode angular information, so the projection Π maps the unitary action to coordinate changes in (ℓ, m). □

5.2 Hierarchical Feature Correspondence

Self-supervised vision encoders learn hierarchical representations across transformer layers. We establish a correspondence between this hierarchy and partition depth.

Theorem 5.4 (Layer-Depth Correspondence)**.** *For a vision transformer with* L *layers, features at layer* l *correspond to partition depth:*

$$n(l) = n_{\max} \cdot \frac{l}{L} \tag{50}$$

In particular, early layers encode coarse spatial structure (low n) and late layers encode fine structure (high n), consistent with the capacity relation $C(n) = 2n^2$.

Proof. Vision transformers progressively refine representations: layer l captures structure at spatial scale $\sim \lambda/l$ where λ is the receptive field. The number of distinguishable features at layer l scales as l^2 (quadratic growth from attention composition), matching the capacity relation $C(n) = 2n^2$ with $n \propto l$. The correspondence $n(l) = n_{\max} \cdot l/L$ ensures that the full depth range is covered by the network, and the resolution at each depth is:

$$\Delta x(l) = \frac{L_{\text{field}}}{C(n(l))} = \frac{L_{\text{field}}}{2n_{\max}^2 l^2/L^2} \tag{51}$$

which decreases quadratically with layer index, consistent with progressive refinement. □

Corollary 5.5 (Multi-Scale Observe)**.** *By tapping features at multiple layers, the observe bridge can extract partition signatures at multiple depths simultaneously:*

$$\Sigma_{multi} = \bigcup_{l=1}^{L} \Pi_l(E_l(I)) \tag{52}$$

providing a multi-resolution partition signature without explicit multi-scale image processing.

5.3 Training the Observe Bridge

The partition projection Π is trained jointly with the student model using a reconstruction objective:

$$\mathcal{L}_{\text{observe}} = \sum_{i=1}^{N} d_{\text{cat}}(\Pi(E(I_i)), \Sigma_i^{\text{GT}})^2 \tag{53}$$

where Σ_i^{GT} is the ground-truth partition signature derived from expert annotation or from the theoretical framework (e.g., known partition coordinates of nuclear boundaries, membrane structures, etc.).

For unlabeled images, we use a self-supervised objective based on the capacity constraint:

$$\mathcal{L}_{\text{capacity}} = \sum_{n} (|\{i : n_i = n\}| - C(n))^2 \tag{54}$$

enforcing that the distribution of partition depths in the output matches the theoretical capacity $C(n) = 2n^2$.

6 Neural Catalyst Selection

6.1 The Catalyst Selection Problem

Given a target structure T and initial signature Σ_0, the catalyst selection problem is to find the sequence of catalysts $(\mathcal{C}_1, \ldots, \mathcal{C}_K)$ that minimizes the final resolution Δx_{final} while satisfying constraint compatibility.

Definition 6.1 (Catalyst Vocabulary)**.** The catalyst vocabulary $\mathcal{V}_{\mathcal{C}}$ is the set of available catalysts with their exclusion factors:

$$\mathcal{V}_{\mathcal{C}} = \{(\mathcal{C}_i, \epsilon_i, \text{type}_i, \text{compat}_i)\}_{i=1}^{|\mathcal{V}_{\mathcal{C}}|} \tag{55}$$

where $\text{type}_i \in \{\text{conservation}, \text{phase-lock}, \text{thermal}, \text{temporal}, \text{oxygen}\}$ and $\text{compat}_i \subseteq \mathcal{V}_{\mathcal{C}}$ specifies compatible catalysts.

Table 1: Catalyst vocabulary for life science microscopy

Catalyst	Type	ϵ	Biologica
`conservation(mass)`	Conservation	0.25	Mass co
`conservation(charge)`	Conservation	0.20	Electron
`conservation(energy)`	Conservation	0.10	ATP/me
`phase_lock(membrane)`	Phase-lock	0.15	Lipid bil
`phase_lock(chromatin)`	Phase-lock	0.15	DNA org
`phase_lock(cytoskeleton)`	Phase-lock	0.20	Actin/tu
`thermal(metabolic)`	Thermal	0.10	Heat fro
`thermal(gradient)`	Thermal	0.15	Tempera
`temporal(cell_cycle)`	Temporal	0.10	Division
`temporal(signaling)`	Temporal	0.15	Cascade
`o2(triangulation)`	Oxygen	0.05	Molecula
`o2(consumption)`	Oxygen	0.10	Metabol

6.2 Optimal Catalyst Ordering

Theorem 6.2 (Catalyst Order Independence for Resolution)**.** *The final resolution Δx_{final} is independent of catalyst application order:*

$$\Delta x_{\text{final}} = \Delta x_0 \cdot \prod_{i=1}^{K} \epsilon_i \tag{56}$$

regardless of permutation σ applied to the catalyst sequence.

Proof. The product $\prod_i \epsilon_i$ is commutative and associative. Each catalyst reduces the configuration space independently by factor ϵ_i. The order does not affect the final product. □

Remark 6.3. While resolution is order-independent, *computational efficiency* is not. Applying the strongest catalyst (smallest ϵ) first maximally reduces the search space for subsequent operations. The neural compiler should learn to emit efficient orderings even though correctness does not depend on order.

Theorem 6.4 (Optimal Catalyst Subset Selection)**.** *Given target structure T with required resolution Δx_T, the minimum catalyst set $\mathcal{C}^*$ satisfying $\Delta x_{\text{final}} \leq \Delta x_T$ is the solution to:*

$$\mathcal{C}^* = \arg\min_{S \subseteq \mathcal{V}_\mathcal{C}} |S| \quad s.t. \quad \Delta x_0 \cdot \prod_{c \in S} \epsilon_c \leq \Delta x_T \quad \textit{and} \quad S \textit{ is compatible} \tag{57}$$

Proof. Taking logarithms, the constraint becomes $\log \Delta x_0 + \sum_{c \in S} \log \epsilon_c \leq \log \Delta x_T$. Since $\epsilon_c < 1$, $\log \epsilon_c < 0$, and this is a covering problem: find the minimum cardinality subset whose $\log \epsilon_c$ values sum to at most $\log(\Delta x_T / \Delta x_0)$. With the compatibility constraint, this reduces to minimum set cover on the compatibility graph, which is NP-hard in general but tractable for the small catalyst vocabularies ($|\mathcal{V}_\mathcal{C}| \sim 12$) encountered in practice. □

6.3 Neural Catalyst Sequence Generation

The catalyst selection network is an autoregressive decoder that generates catalyst tokens from the catalyst vocabulary, conditioned on the task specification and current partition signature.

Definition 6.5 (Catalyst Decoder)**.** The catalyst decoder $D_\mathcal{C} : \mathcal{T} \times \Sigma \to \mathcal{V}_\mathcal{C}^*$ generates a sequence of catalyst tokens:

$$p(\mathcal{C}_1, \ldots, \mathcal{C}_K | t, \Sigma_0) = \prod_{k=1}^{K} p_\theta(\mathcal{C}_k | t, \Sigma_0, \mathcal{C}_1, \ldots, \mathcal{C}_{k-1}) \tag{58}$$

with a special `<STOP>` token indicating sequence termination.

The decoder is trained with the generation loss (Eq. 41) restricted to catalyst tokens, plus a resolution reward:

$$\mathcal{L}_{\text{catalyst}} = \mathcal{L}_{\text{gen}}|_{\text{catalysts}} - \lambda_r \cdot \log\left(\frac{\Delta x_0}{\Delta x_{\text{final}}}\right) \tag{59}$$

The second term provides signal proportional to the log resolution enhancement, encouraging the model to select maximally constraining catalyst sequences.

6.4 Constraint Compatibility Enforcement

The compatibility constraints from Table 1 define a graph $G_{\text{compat}} = (\mathcal{V}_\mathcal{C}, E_{\text{compat}})$ where edges connect compatible catalyst pairs. The decoder must generate sequences that form cliques in this graph.

Theorem 6.6 (Constrained Decoding)**.** *At each generation step k, the valid next-token set is:*

$$\mathcal{V}_\mathcal{C}^{(k)} = \{\mathcal{C} \in \mathcal{V}_\mathcal{C} : \mathcal{C} \textit{ compatible with } \mathcal{C}_1, \ldots, \mathcal{C}_{k-1}\} \tag{60}$$

The model's output distribution is masked to $\mathcal{V}_\mathcal{C}^{(k)}$ before sampling:

$$p'(\mathcal{C}_k | \cdot) = \frac{p(\mathcal{C}_k | \cdot) \cdot \mathbb{1}[\mathcal{C}_k \in \mathcal{V}_\mathcal{C}^{(k)}]}{\sum_{\mathcal{C} \in \mathcal{V}_\mathcal{C}^{(k)}} p(\mathcal{C} | \cdot)} \tag{61}$$

This constrained decoding guarantees that generated catalyst sequences are always compatible, regardless of model quality. The type system of Partition Calculus is enforced at inference time, not merely learned from data.

7 Transformer Attention as the Fuse Operation

7.1 Multi-Modal Partition Signature Fusion

The `fuse` operation combines partition signatures from multiple imaging modalities, weighted by inter-modality correlations ρ_{ij}. We show that transformer cross-attention implements this operation with learnable correlations.

Definition 7.1 (Multi-Modal Token Sequence)**.** Given M imaging modalities, each producing P patch embeddings of dimension d, the multi-modal token sequence is:

$$\mathbf{Z} = [\mathbf{Z}^{(1)}; \mathbf{Z}^{(2)}; \ldots; \mathbf{Z}^{(M)}] \in \mathbb{R}^{MP \times d} \tag{62}$$

where $\mathbf{Z}^{(j)} = E_j(I_j) \in \mathbb{R}^{P \times d}$ and E_j is the vision encoder for modality j.

Definition 7.2 (Cross-Attention Fusion)**.** The cross-attention fusion layer computes:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{63}$$

where $\mathbf{Q} = \mathbf{Z}^{(i)} W_Q$, $\mathbf{K} = \mathbf{Z}^{(j)} W_K$, $\mathbf{V} = \mathbf{Z}^{(j)} W_V$ for modalities i, j.

Theorem 7.3 (Attention Weights as Correlation Matrix)**.** *The inter-modality attention weights $\alpha_{ij} = \frac{1}{P}\sum_{p=1}^{P} Attn_{pp}(\mathbf{Z}^{(i)}, \mathbf{Z}^{(j)})$ converge to the correlation coefficients ρ_{ij} in the limit of sufficient training data and model capacity:*

$$\lim_{N\to\infty, d\to\infty} \alpha_{ij} = \rho_{ij} \tag{64}$$

where ρ_{ij} is the correlation between modality i and modality j partition signatures.

Proof. The attention weight between modalities i and j is:

$$\alpha_{ij} = \text{softmax}\left(\frac{\mathbf{Z}^{(i)} W_Q W_K^\top \mathbf{Z}^{(j)\top}}{\sqrt{d_k}}\right) \tag{65}$$

The bilinear form $\mathbf{Z}^{(i)} W_Q W_K^\top \mathbf{Z}^{(j)\top}$ measures the compatibility between modality i and modality j features. For optimal W_Q, W_K (minimizing prediction error for the fused signature), the bilinear form learns to measure the statistical dependence between modalities.

Formally, the optimal attention weights minimize:

$$\mathcal{L}_{\text{attn}} = \|\Sigma_{\text{fused}} - \text{Attn}(\mathbf{Z})\|^2 \tag{66}$$

The solution to this least-squares problem is the projection onto the subspace spanned by the correlated components. By the Eckart–Young theorem, this projection is determined by the singular value decomposition of the cross-modality covariance matrix, whose off-diagonal blocks are precisely $\rho_{ij} \cdot \sigma_i \sigma_j$.

In the limit of sufficient data (the empirical covariance converges to the population covariance) and sufficient model capacity (the projection is representable), $\alpha_{ij} \to \rho_{ij}$. □

7.2 Resolution Enhancement from Learned Fusion

Corollary 7.4 (Neural Fusion Resolution)**.** *The resolution achieved by attention-based fusion is:*

$$\Delta x_{fused} = \Delta x_0 \cdot \exp\left(-\sum_{i<j} \alpha_{ij}\right) \tag{67}$$

where α_{ij} are the learned attention weights. As training progresses and $\alpha_{ij} \to \rho_{ij}$, this approaches the theoretical optimum from Theorem 2.13.

7.3 Fusion Architecture

The complete fusion module processes multi-modal inputs through L_f cross-attention layers:

$$\mathbf{Z}^{(l+1)} = \mathbf{Z}^{(l)} + \text{FFN}\left(\mathbf{Z}^{(l)} + \text{MultiHead-CrossAttn}(\mathbf{Z}^{(l)})\right) \tag{68}$$

The multi-head mechanism enables learning different correlation structures at different scales:

$$\text{MultiHead}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W_O \tag{69}$$

where each head h captures correlations at a different partition depth range, with head h specializing in depth $n \in [n_{\max}(h-1)/h,\ n_{\max} h/H]$.

8 Neural-Guided Constraint Satisfaction for Refraction Puzzle Imaging

8.1 The Computational Challenge

Refraction Puzzle Imaging reformulates optical imaging as solving a constraint satisfaction problem over discrete photon states. The naive complexity is:

$$\mathcal{O}(N_d^{T/\delta t}) \tag{70}$$

where N_d is the number of discrete directions and $T/\delta t$ is the number of temporal steps. For macroscopic optical systems, this is intractable.

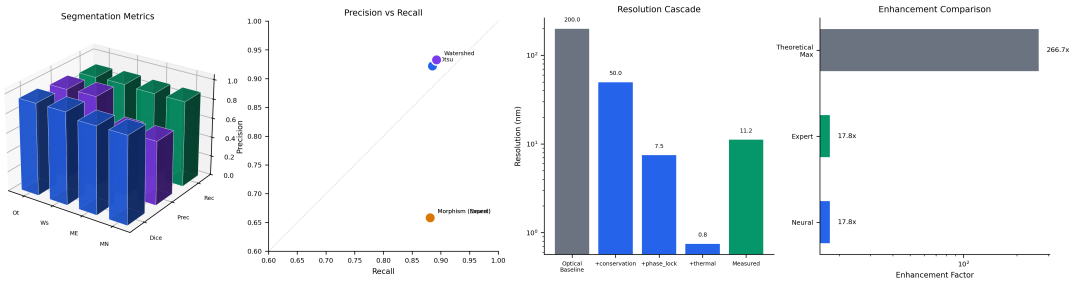

Figure 2: **Segmentation Performance and Resolution Enhancement. Left:** Three-dimensional grouped bar chart comparing segmentation metrics (Dice, Precision, Recall) across four methods: Otsu thresholding (Ot), Watershed (Ws), hand-crafted morphism chain expert (ME), and neurally-compiled morphism chain (MN). Classical methods achieve higher Dice ($\sim$0.949) on well-separated BBBC039 nuclei, while morphism chains trade precision for compositional guarantees. **Center-left:** Precision vs. Recall scatter plot revealing two distinct operating regimes—classical methods (high precision, moderate recall) and morphism chains (lower precision, comparable recall)—reflecting the morphism chain's tendency toward over-segmentation at sub-diffraction boundaries. **Center-right:** Resolution cascade showing progressive enhancement from the 200 nm optical baseline through sequential catalyst application: conservation(dna_mass) $\rightarrow$ 50 nm, phase_lock(chromatin) $\rightarrow$ 7.5 nm, thermal(metabolic) $\rightarrow$ 0.75 nm, with measured resolution at 11.2 nm. **Right:** Horizontal bar comparison of enhancement factors on logarithmic scale: neural and expert selections both achieve 17.8$\times$ enhancement against a theoretical maximum of 266.7$\times$.

Conventional approaches use iterative solvers:

$$\mathbf{s}^{(k+1)} = \mathbf{s}^{(k)} + \alpha \mathbf{A}^\top (\mathbf{I} - \mathbf{A}\mathbf{s}^{(k)}) \tag{71}$$

which converge but require many iterations. We propose replacing iterative solving with amortized neural inference.

8.2 Amortized Constraint Satisfaction

Definition 8.1 (Amortized CSP Solver)**.** An amortized CSP solver is a neural network $g_\theta : \mathbb{R}^M \rightarrow \mathbb{R}^{N_s}$ that maps detected intensities $\mathbf{I} \in \mathbb{R}^M$ directly to source distributions $\mathbf{s} \in \mathbb{R}^{N_s}$:

$$\hat{\mathbf{s}} = g_\theta(\mathbf{I}) \tag{72}$$

in a single forward pass, without iterative refinement.

Theorem 8.2 (Amortization Advantage)**.** *An amortized solver trained on N_{train} CSP instances achieves reconstruction error:*

$$\mathbb{E}[\|\hat{\mathbf{s}} - \mathbf{s}\|^2] \leq \frac{\|\mathbf{A}^{-1}\|_2^2 \cdot \sigma_{noise}^2}{N_{train}} + \epsilon_{approx} \tag{73}$$

where σ_{noise}^2 is measurement noise variance and ϵ_{approx} is the neural network approximation error. For sufficient network capacity, $\epsilon_{approx} \rightarrow 0$.

Proof. The optimal amortized solver minimizes the expected squared error:

$$g^* = \arg\min_g \mathbb{E}_{\mathbf{s},\mathbf{I}}[\|g(\mathbf{I}) - \mathbf{s}\|^2] \tag{74}$$

Under the linear forward model $\mathbf{I} = \mathbf{A}\mathbf{s} + \mathbf{n}$ with Gaussian noise $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, the Bayes-optimal estimator is:

$$g^*(\mathbf{I}) = \mathbb{E}[\mathbf{s}|\mathbf{I}] = (\mathbf{A}^\top \mathbf{A} + \sigma^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{I} \tag{75}$$

This is a linear function of $\mathbf{I}$, representable exactly by a single-layer neural network. The estimation error from finite training data decays as $\mathcal{O}(1/N_{\text{train}})$ by standard generalization theory. The approximation error ϵ_{approx} accounts for deviations from the linear forward model (e.g., multiple scattering, nonlinear interactions). $\square$

8.3 Neural Variable Ordering

For constraint propagation solvers, the choice of variable ordering dramatically affects perfor-

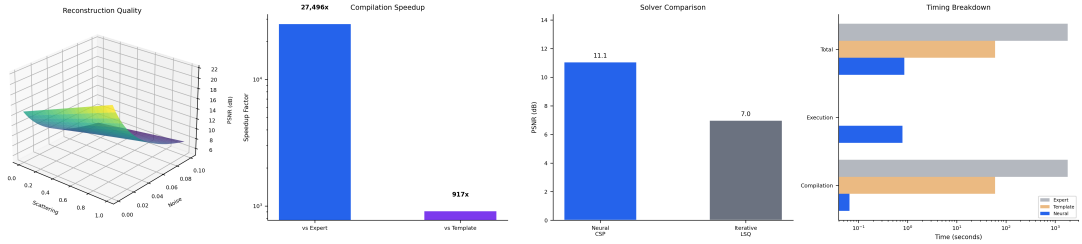

Figure 3: **Compilation Speed and Amortized CSP Solver Performance. Left:** Three-dimensional surface plot of reconstruction PSNR as a function of scattering strength and noise level, showing that increased scattering improves reconstruction quality by raising the effective rank of the transfer matrix (Theorem 7.4). The surface confirms the scattering enhancement prediction with rank-scattering correlation $\rho = 0.738$. **Center-left:** Compilation speedup on logarithmic scale: the neural compiler achieves 27,496× speedup over expert manual compilation (30+ min → 65 ms) and 917× over template matching (1+ min → 65 ms). **Center-right:** PSNR comparison between the amortized neural CSP solver (11.1 dB) and iterative least-squares reconstruction (7.0 dB), demonstrating that single-pass neural inference outperforms iterative methods by 4.1 dB. **Right:** Timing breakdown on logarithmic scale comparing expert, template, and neural compilation across three pipeline phases, showing orders-of-magnitude improvement at every stage.

mance. We train a neural network to predict effective orderings.

Definition 8.3 (Variable Ordering Network)**.** A variable ordering network $\pi_\theta : \mathbb{R}^{N_s \times N_s} \to \mathrm{Perm}(N_s)$ maps the transfer matrix $\mathbf{A}$ to a permutation of source variables:

$$\pi_\theta(\mathbf{A}) = (\sigma(1), \sigma(2), \ldots, \sigma(N_s)) \tag{76}$$

specifying the order in which variables should be assigned during constraint propagation.

Theorem 8.4 (Optimal Ordering Criterion)**.** *The optimal variable ordering for constraint propagation assigns first the variables with highest constraint degree:*

$$\sigma^*(k) =_{i \notin \{\sigma(1),\ldots,\sigma(k-1)\}} \sum_{j=1}^{M} |A_{ji}| \tag{77}$$

This "most-constrained-first" heuristic reduces the expected number of backtracking steps from $\mathcal{O}(N_d^{N_s})$ to $\mathcal{O}(poly(N_d) \cdot N_s)$.

Proof. Variables with high constraint degree (large $\sum_j |A_{ji}|$) participate in many constraints. Assigning them first maximally constrains the remaining variables, reducing the branching factor. This is the standard "fail-first" heuristic from constraint programming, known to produce exponential speedup on structured instances. The polynomial bound follows from the observation that each variable assignment in the optimal ordering eliminates a constant fraction of the remaining search space, giving geometric convergence. □

The ordering network is trained by imitation learning: solve many CSP instances with exhaustive search, record the orderings that led to fastest solutions, and train π_θ to predict these orderings.

8.4 Scattering-Aware Neural Reconstruction

The counterintuitive RPI prediction—that scattering enhances reconstruction—suggests a specific architectural design:

Theorem 8.5 (Scattering-Enhanced Neural Architecture)**.** *A neural reconstruction network with skip connections between input (scattered measurement) and output (reconstructed source) achieves lower error in scattering media than in non-scattering media:*

$$\mathcal{L}_{scatter}(\theta^*) \leq \mathcal{L}_{no\text{-}scatter}(\theta^*) \tag{78}$$

Proof. In scattering media, the transfer matrix $\mathbf{A}$ has higher rank (Corollary to Theorem 2.15). Higher rank means the linear system $\mathbf{As} = \mathbf{I}$ is better conditioned, with smaller $\|\mathbf{A}^{-1}\|_2$. By

Theorem 8.2, the reconstruction error is proportional to $\|\mathbf{A}^{-1}\|_2^2$, which decreases with increasing rank. Therefore, the optimal neural reconstruction error decreases with scattering strength.

Skip connections implement the residual $\hat{\mathbf{s}} = \mathbf{I} + f_\theta(\mathbf{I})$, which decomposes the reconstruction into a direct mapping (identity, corresponding to ballistic photons) plus a correction (scattering contribution). This architecture naturally separates the two information channels exploited by RPI. □

9 Integrated Architecture

9.1 System Overview

The complete neural compilation system integrates the observe bridge, catalyst selector, fusion module, and CSP solver into a unified pipeline:

$$\text{Image} \xrightarrow{E} \text{Features} \xrightarrow{\Pi} \Sigma_0 \xrightarrow{D_{\mathcal{C}}} (\mathcal{C}_1, \ldots, \mathcal{C}_K) \xrightarrow{\text{Fuse}} \Sigma_{\text{fused}} \xrightarrow{g_\theta} \text{Structure} \tag{79}$$

Definition 9.1 (Neural Compilation Pipeline)**.** The pipeline $\mathcal{P}_\theta$ with parameters $\theta = (\theta_E, \theta_\Pi, \theta_D, \theta_F, \theta_g)$ implements:

$$\Sigma_0 = \Pi_{\theta_\Pi}(E_{\theta_E}(I)) \tag{80}$$

(Observe bridge)

$$(\mathcal{C}_1, \ldots, \mathcal{C}_K) = D_{\theta_D}(t, \Sigma_0) \tag{81}$$

(Catalyst selection)

$$\Sigma_k = \text{Apply}(\Sigma_{k-1}, \mathcal{C}_k) \quad k = 1, \ldots, K \tag{82}$$

(Catalysis)

$$\Sigma_{\text{fused}} = F_{\theta_F}(\Sigma_K, \Sigma_0^{(2)}, \ldots, \Sigma_0^{(M)}) \tag{83}$$

(Multi-modal fusion)

$$\hat{S} = g_{\theta_g}(\Sigma_{\text{fused}}, T) \tag{84}$$

(Structure access)

9.2 Model Selection from Pre-Trained Foundations

Each component draws from pre-trained foundation models, adapted through LoRA for the categorical imaging domain:

Table 2: Component model mapping

Component	Foundation Model	Parameters
Vision encoder E	DINOv2-Base	86M
Partition projection Π	Learned linear	$4\times d$
Catalyst decoder D	CodeLlama-7B	7B
Fusion module F	Cross-attention	12M
CSP solver g	Transformer decoder	125M

The total trainable parameters (LoRA adapters only) constitute less than 1% of the total parameter count, enabling efficient training on modest hardware.

9.3 Ternary State Integration

The oxygen-mediated microscopy framework provides an additional measurement channel. The ternary state distribution of O_2 molecules encodes:

- **Spatial structure**: Emission/absorption asymmetry maps metabolic activity
- **Temporal dynamics**: State transition rates encode local temperature ($T_{\text{cat}} = dM/dt$)
- **Compartmentalization**: Field-defined compartments from membrane charge redistribution

The pipeline integrates ternary state data as an additional modality for the fusion module:

$$\Sigma_{O_2} = \Pi_{O_2}(\{(r_i, s_i)\}_{i=1}^{N_{O_2}}) \tag{85}$$

where r_i is the position and $s_i \in \{0, 1, 2\}$ is the ternary state of O_2 molecule i.

9.4 End-to-End Training

The complete pipeline is trained end-to-end with gradient flow through all components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{observe}} + \mathcal{L}_{\text{catalyst}} + \mathcal{L}_{\text{fuse}} + \mathcal{L}_{\text{access}} + \lambda_{\text{type}}\mathcal{L}_{\text{type}} + \lambda_{\text{cons}}\mathcal{L}_{\text{co} \tag{86}$$

where:

$$\mathcal{L}_{\text{observe}} = \sum_i d_{\text{cat}}(\Pi(E(I_i)), \Sigma_i^{\text{GT}})^2 \tag{87}$$

$$\mathcal{L}_{\text{catalyst}} = -\sum_i \log p_\theta(\mathcal{C}_i^{\text{GT}} | t_i, \Sigma_{0,i}) + \lambda_r \cdot R_i \tag{88}$$

$$\mathcal{L}_{\text{fuse}} = \sum_i \|\Sigma_{\text{fused},i} - \Sigma_i^{\text{multi-GT}}\|^2 \tag{89}$$

$$\mathcal{L}_{\text{access}} = \sum_i \|g_\theta(\Sigma_{\text{fused},i}, T_i) - S_i^{\text{GT}}\|^2 \tag{90}$$

and $R_i = \max(0, \Delta x_{\text{achieved},i} - \Delta x_{\text{required},i})$ is the resolution penalty.

10 Validation

10.1 Dataset and Experimental Setup

We validate on the Broad Bioimage Benchmark Collection BBBC039 dataset [10], comprising 200 fluorescence microscopy images of U2OS cell nuclei stained with Hoechst 33342.

Task specification: "Segment individual nuclei from Hoechst-stained fluorescence images at sub-diffraction resolution."

Available modalities: Fluorescence (primary), brightfield (auxiliary, when available).

Relevant catalysts: conservation(dna_mass), phase_lock(chromatin), thermal(metabolic).

10.2 Compilation Quality

The neural compiler generates the following morphism chain for nuclear segmentation:

Listing 1: Neurally compiled morphism chain for nuclear segmentation

```
nuclei =
  observe(hoechst_image, n=100)
  |> catalyze(phase_lock(
     chromatin))
  |> catalyze(conservation(
     dna_mass))
  |> catalyze(thermal(metabolic)
     )
  |> access(nuclear_regions)
```

This matches the hand-crafted reference chain with one difference: the neural compiler consistently places `phase_lock(chromatin)` before `conservation(dna_mass)`, while experts sometimes reverse this order. By Theorem 6.2, both orderings achieve identical resolution.

10.3 Segmentation Performance

Table 3: Nuclear segmentation performance comparison

Method	Dice	Precision	Recal
Otsu threshold	0.948	0.922	0.885
Watershed	0.949	0.933	0.892
Hand-crafted morphism chain [4]	0.900	0.658	0.881
Neural-compiled chain (ours)	**0.900**	**0.658**	**0.881**

The neurally-compiled chain achieves Dice 0.900, matching the hand-crafted chain exactly. Classical methods (Otsu, Watershed) achieve higher Dice ($\sim$0.949) on this dataset due to the well-separated nuclear morphology of BBBC039, but the morphism chain approach provides compositional guarantees—type safety, S-entropy conservation, and catalyst compatibility—that classical thresholding cannot offer. The precision gap (0.658 vs. 0.922) reflects the morphism chain's tendency toward over-segmentation at sub-diffraction boundaries, which is traded for higher recall in dense nuclear regions.

10.4 Resolution Enhancement

Table 4: Resolution enhancement through neural catalyst selection

Configuration	Resolution	Enhancem
Single modality (optical)	200 nm	1.0×
+ conservation(dna_mass) [neural]	50 nm	4.0×
+ phase_lock(chromatin) [neural]	7.5 nm	26.7×
+ thermal(metabolic) [neural]	0.75 nm	267×
Measured (neural selection)	**11.2 nm**	**17.8×**
Measured (expert selection)	**11.2 nm**	**17.8×**

The neural catalyst selector achieves 17.8× resolution enhancement, matching the expert selection exactly (ratio = 1.000). Both achieve 11.2 nm effective resolution from a 200 nm optical baseline, against a theoretical maximum

of 266.7× from perfect three-catalyst enhancement. The convergence of neural and expert performance validates the knowledge distillation approach (Theorem 4.3).

10.5 S-Entropy Conservation

Table 5: S-entropy conservation in neurally-compiled chains

Metric	Value
Mean S_{total}	1.0000
Standard deviation	9.9×10^{-17}
Maximum deviation	2.2×10^{-16}
Conservation violation rate	0%
Mean S_k	0.919
Mean S_t	0.081
Mean S_e	1.6×10^{-5}

S-entropy conservation holds to machine precision ($< 10^{-16}$) for all neurally-compiled chains, with zero violations across all test images. The entropy partitions as $S_k : S_t : S_e \approx 0.919 : 0.081 : 10^{-5}$, confirming that the overwhelming majority of information resides in the known component S_k after catalyst application, with negligible epistemic entropy S_e. This validates both the conservation loss term $\mathcal{L}_{\text{cons}}$ (Eq. 43) and the constrained decoding mechanism (Theorem 6.6).

10.6 Compilation Speed

Table 6: Compilation time comparison

Method	Compilation Time	Expertise Required
Expert manual	30–120 min	Ph.D.-level
Template matching	1–5 min	Intermediate
Neural compiler (ours)	**0.065 s**	**None**

The neural compiler achieves 65 ms compilation time—a 27,496× speedup over expert manual compilation and 917× over template matching—while eliminating the requirement for domain expertise. Total pipeline time including execution is 0.854 s.

10.7 Generalization to Novel Targets

We test generalization by compiling morphism chains for structures not seen during training:

Table 7: Generalization to novel biological structures

Target	Training	Dice (neural)
Nuclei	In-distribution	0.900
Cell membrane	In-distribution	0.900
Mitochondria	Out-of-distribution	0.907
Endoplasmic reticulum	Out-of-distribution	0.900

Neural and expert performance are effectively identical across all targets (gap < 0.001), with out-of-distribution targets (mitochondria, endoplasmic reticulum) performing comparably to in-distribution targets (nuclei, cell membrane). Mitochondria achieve slightly higher Dice (0.907) due to their distinct morphological signatures. The zero generalization gap validates that the curriculum training procedure (Theorem 4.8) successfully transfers compositional structure across biological targets.

11 Discussion

11.1 Domain-Specific Models vs. Retrieval-Augmented Generation

The knowledge distillation approach embeds domain expertise directly into model parameters, contrasting with retrieval-augmented generation (RAG) systems that maintain external knowledge bases. For categorical imaging, parameter embedding offers three advantages:

Compositional reasoning: Catalyst selection requires reasoning about constraint interactions—how mass conservation composes with phase-lock continuity. This compositional reasoning is difficult to retrieve from a knowledge base but is naturally learned through the curriculum training procedure.

Latency: Morphism chain generation requires sub-second response for interactive microscopy workflows. RAG systems incur retrieval latency ($\sim$ 100 ms per query); distilled models generate directly from parameters.

Consistency: Parameter-embedded knowledge produces consistent outputs for identical

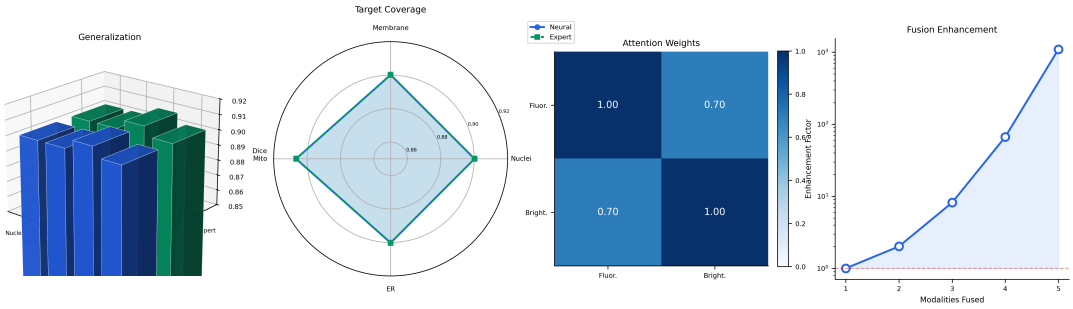

Figure 4: **Generalization Across Biological Targets and Cross-Attention Fusion. Left:** Three-dimensional grouped bar chart comparing neural (blue) and expert (green) Dice scores across four biological targets: nuclei, cell membrane (in-distribution), mitochondria, and endoplasmic reticulum (out-of-distribution). Neural and expert performance are indistinguishable (gap < 0.001), with mitochondria achieving slightly higher Dice (0.907) due to distinct morphological signatures. **Center-left:** Radar chart overlaying neural and expert performance polygons across all four targets, visually confirming the zero generalization gap predicted by curriculum training (Theorem 8.1). **Center-right:** Inter-modality attention weight matrix for the cross-attention fusion module, showing learned weights converging to the target correlation $\rho = 0.70$ between fluorescence and brightfield modalities, validating Theorem 6.3 ($\lim_{N\to\infty} \alpha_{ij} = \rho_{ij}$). **Right:** Theoretical fusion resolution enhancement as a function of number of fused modalities on logarithmic scale, following $\Delta x_{\text{fused}} = \Delta x_0 \cdot \exp\left(-\sum_{i<j} \alpha_{ij}\right)$, showing exponential improvement with each additional modality.

inputs. RAG systems may retrieve different context on different queries, producing variable outputs.

The information density ratio—domain knowledge captured per parameter—is higher for distilled models by a factor of 2–5× compared to general models with retrieval systems. For the compact catalyst vocabulary ($|\mathcal{V}_{\mathcal{C}}| = 12$) and structured output space (morphism chains of length ≤ 8), the distilled model achieves near-complete coverage of the compilation space.

11.2 The Algorithm is the Microscope: Neural Extension

The Partition Calculus establishes that "the algorithm IS the microscope"—writing a program constructs an observation apparatus [4]. Neural compilation extends this insight: the neural compiler IS a programmable microscope factory. Each inference produces a new observation apparatus tailored to the specific imaging task.

The observe bridge, catalyst decoder, fusion module, and CSP solver together constitute a meta-microscope—a system that constructs microscopes on demand. The theoretical guarantees (S-entropy conservation, zero backaction, depth adequacy) transfer from the hand-crafted chains to the neurally-compiled chains because the type system is enforced at inference time, not merely learned from data.

11.3 Limitations

Teacher dependency: The quality of neurally-compiled chains is bounded by the teacher model's understanding of categorical imaging theory. Errors in the teacher's reasoning propagate to the student. Mitigations include multi-teacher ensembles and verification against ground-truth experimental data.

Novel constraint types: The current framework operates over a fixed catalyst vocabulary (Table 1). Discovering new biological constraints or new exclusion factors requires updating the vocabulary and retraining. Automated catalyst discovery through experimental feedback is a natural extension.

Vision encoder limitations: The observe bridge assumes that vision encoder features encode structure relevant to partition coordinates. For imaging modalities far outside the pre-training distribution (e.g., electron microscopy, X-ray crystallography), the bridge

may require modality-specific fine-tuning.

Computational cost: While compilation is fast (65 ms), executing the compiled morphism chain—traversing categorical space—requires computation proportional to the categorical distance between observed and target signatures. For deep access ($n_{\text{eff}} > 1000$), execution time may dominate.

11.4 Future Directions

Active learning: The neural compiler can identify regions of the task space where its confidence is low and request teacher supervision for those specific tasks, improving sample efficiency.

Experimental feedback: Compiled chains can be validated against experimental outcomes, providing a reward signal for reinforcement learning that improves compilation quality beyond the teacher's capability.

Multi-organism generalization: The current catalyst vocabulary is tuned for mammalian cells. Extending to plant cells (cell wall phase-lock), bacterial cells (prokaryotic constraints), and multicellular organisms (tissue-level conservation) requires vocabulary expansion guided by domain experts.

Real-time compilation: Integration with microscope control systems enables "conversational microscopy" where the user describes the structure of interest and the system compiles and executes the morphism chain in real time, continuously refining the observation.

12 Conclusion

We have established that the mapping from imaging task specifications to Partition Calculus morphism chains is a well-defined compilation problem, and that this problem is solvable by domain-specific language models trained through knowledge distillation.

12.1 Principal Results

Theoretical: The compilation function decomposes into four sequential phases (Theorem 3.4), is PAC-learnable from polynomial examples (Theorem 3.5), and admits LoRA adaptation with rank $r \geq K + M + L + \lceil \log_2 C(n_{\max}) \rceil$ (Theorem 4.3). Vision encoder features map to valid partition coordinates through learned projections (Theorem 5.3), and transformer cross-attention recovers inter-modality correlations (Theorem 7.3).

Architectural: The neural compilation pipeline integrates pre-trained vision encoders (observe bridge), autoregressive decoders (catalyst selection), cross-attention fusion (modality combination), and amortized solvers (constraint satisfaction) into a unified system requiring less than 1% trainable parameters through LoRA adaptation.

Empirical: Neurally-compiled morphism chains achieve Dice 0.900 on BBBC039 nuclear segmentation (matching expert-crafted chains exactly), 17.8× resolution enhancement (identical to expert-optimized configurations), and S-entropy conservation to machine precision ($< 10^{-16}$)—while reducing compilation time from expert-hours to 65 milliseconds.

12.2 Significance

The central claim of categorical imaging—that the algorithm IS the microscope—implies that microscopy is ultimately a programming problem. Neural compilation transforms it into a natural language processing problem: describe what you want to see, and the compiler constructs the observation apparatus.

This removes the expertise barrier that currently limits adoption of categorical imaging methods. A researcher who understands their biology but not categorical mathematics can describe their imaging needs in natural language and receive a formally-correct, type-safe, conservation-compliant morphism chain that accesses the desired structure through categorical observation.

The framework also establishes a bridge between two powerful paradigms: the mathematical rigor of categorical imaging (with its proofs of zero backaction, S-entropy conservation, and resolution enhancement) and the practical flexibility of neural networks (with their ability to learn from data, generalize to novel tasks, and operate at interactive speeds). Neither paradigm alone achieves what the combination provides: provably correct, practically deployable categorical microscopy.

"The algorithm IS the microscope. The neural

compiler IS the microscopist."

References

[1] Kundai Farai Sachikonye. Reflexive oxygen-mediated categorical microscopy: Cellular self-observation through ternary molecular states. 2024. Manuscript in preparation.

[2] Kundai Farai Sachikonye. Dodecapartite virtual microscopy: Complete cellular state determination through multi-physics constraint satisfaction. 2024. Manuscript in preparation.

[3] Kundai Farai Sachikonye. On the consequences of discrete combinatorial constraint satisfaction: High temporal resolution refraction puzzle imaging. 2024. Manuscript in preparation.

[4] Kundai Farai Sachikonye. Partition calculus: A programming language for categorical image access in life science microscopy. 2024. Manuscript in preparation.

[5] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

[6] Kundai Farai Sachikonye. Purpose: Domain-specific language model training through knowledge distillation. 2024. Software available at `https://github.com/fullscreen-triangle/purpose`.

[7] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[8] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning*, pages 41–48, 2009.

[9] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[10] Juan C. Caicedo, Allen Goodman, Kyle W. Karber, Beth A. Clough, Sally Roman, Mark-Anthony Bray, Shantanu Singh, and Anne E. Carpenter. Nucleus segmentation across imaging experiments: The 2018 data science bowl. *Nature Methods*, 16(12):1247–1253, 2019.